

Weakly-supervised localisation generalises where genotype prediction does not: an externally validated null for EGFR radiogenomics from lung CT

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Abstract

Background. CT-based prediction of *EGFR* mutation status is widely reported at AUC 0.80–0.89, almost always from single-centre cohorts of 200–800 patients without external validation.

Purpose. To evaluate whether a grounded vision–language model predicts *EGFR* status from lung CT beyond routinely available clinical variables, and whether its supervised attention localises tumours on an unseen cohort.

Materials and Methods. Retrospective secondary analysis of the public NSCLC-Radiogenomics collection (211 subjects; 153 with a known *EGFR* call, 40 mutant, 26.1%). A BiomedCLIP ViT-B/16 encoder with a region-query grounding head supervised against tumour segmentations was trained under nested five-fold cross-validation with three seeds. The primary endpoint was external generalisation of localisation, evaluated on 420 held-out NSCLC-Radiomics patients against three controls: a per-fold spatial shuffle, a centre prior, and a randomly initialised head. A co-primary endpoint — the incremental AUC over a five-variable clinical model — was pre-declared as an expected null. The analysis plan was fixed before external evaluation.

Results. Pooled out-of-fold *EGFR* AUC was 0.618 ± 0.025 , against 0.794 for smoking status alone. Adding the model to clinical variables changed AUC by -0.029 ± 0.015 , every seed’s confidence interval including zero; no fusion topology exceeded the clinical model. Calibration slope was 0.19 and the Brier score (0.236) exceeded the base-rate floor (0.193). Localisation transferred: attention-mass lift $+0.244$ over the shuffle in 11 of 15 external evaluations, pointing 0.609 versus 0.476 for a centre prior and 0.005 for a randomly initialised head.

Conclusion. Supervised localisation generalises across institutions while lesion-anchored *EGFR* prediction does not exceed a five-variable chart model.

Abbreviations: AUC = area under the receiver operating characteristic curve; CV = cross-validation; IQR = interquartile range; NSCLC = non-small cell lung cancer; TKI = tyrosine kinase inhibitor.

1 Introduction

EGFR genotype directs tyrosine kinase inhibitor therapy in non-small cell lung cancer. Tissue is often limited and molecular testing is not universally available, so a non-invasive imaging surrogate would have real clinical value. This has motivated a large radiogenomics literature.

That literature reports *EGFR* prediction from CT at AUC 0.80–0.89 [5, 6, 8–10]. The cohorts are typically 200–826 patients, usually single-centre, and external validation is uncommon. Against

this, the one large study with external validation — 1,646 patients and 11,473 segmented lesions with next-generation-sequencing labels — reports internal AUC 0.62–0.68 and external 0.55–0.63 [4]. Notably, its external cohort is NSCLC-Radiogenomics, the collection used here.

That discrepancy is this study’s motivation. We do not propose a better classifier. We separate two claims that are routinely bundled — that such a model *localises* the lesion, and that it *genotypes* it — and test each with the controls it requires, on a cohort where the second can be measured against the clinical variables it would have to beat.

2 Materials and Methods

2.1 Study design and cohorts

This retrospective secondary analysis used publicly available, de-identified imaging and did not constitute human-subjects research. The analysis plan, including the primary endpoint and all controls, was fixed and version-controlled before external evaluation; deviations are logged in the public repository.

The internal cohort is NSCLC-Radiogenomics [1], a Stanford University / VA Palo Alto series distributed by The Cancer Imaging Archive. Participant flow is given in Figure 1: 211 subjects in the clinical sheet, 188 with an imaging series, 158 successfully preprocessed, 153 with a known *EGFR* call, 133 adenocarcinoma, and 97 with both adenocarcinoma histology and a pixel-level segmentation. That last set is the pre-specified primary analysis set. Cohort characteristics appear in Table 1.

No accrual window is reported. The collection’s **CT Date** values span 1989–1996 and are de-identification offsets rather than dates; publishing a range derived from them would misdescribe the cohort. The index-test-to-reference- standard interval, which survives the offset because it is a difference, has a median of 38 days (IQR 19.5–65.5).

The external cohort is NSCLC-Radiomics [3], 420 preprocessed patients from Maastric Clinic (Netherlands) — a different country, scanner fleet, and clinical population (radiotherapy planning rather than surgical). The two cohorts share no patients, and no evaluated checkpoint was warm-started from the external collection; both facts are asserted in the repository’s test suite rather than claimed in prose.

2.2 Model

A BiomedCLIP ViT-B/16 encoder (last two blocks unfrozen) produces patch tokens per slice, aggregated across 16 slices by a masked mean. A grounding head applies eight learned region queries by cross-attention, and the resulting attention maps are supervised against the tumour region of interest projected onto the patch grid. The classifier pools from the grounded region embeddings, so prediction and localisation share a representation.

Inputs are axial crops of $(\text{bbox}_{\text{mm}} + 24) \times 2.5$ around the lesion with 30% positional jitter, windowed at centre -600 HU / width 1500 HU and resized to 224×224 .

2.3 Endpoints, controls, and statistics

H₁ (primary). External attention-mass lift over a per-fold spatial shuffle is greater than zero. Three controls are mandatory in every reported table:

- **Spatial shuffle** — the same attention values permuted across cells, isolating *where* mass fell from *how peaked* the map is.

- **Centre prior** — external ROI centroids sit at (0.50, 0.51) of the frame (sd $\approx$ 0.08) covering 5.8% of area, so “look at the middle” is a strong baseline. Pointing and area-matched IoU are invariant to the prior’s width, so that margin cannot be tuned.
- **Weight randomisation** — an untrained head of identical architecture, ruling out architectural inductive bias.

H₂ (co-primary, pre-declared null). $\Delta\text{AUC} = \text{AUC}(\text{clinical} + \text{model}) - \text{AUC}(\text{clinical})$.

Evaluation used nested five-fold cross-validation stratified on *EGFR*, three seeds, with within-fold rank normalisation before pooling. The headline is the across-seed mean $\pm$ standard deviation; a pooled confidence interval absorbs no rerun variance and is never quoted alone. **No operating point is defined:** the index test is continuous and the calibration reported in Section 3.2 does not support a threshold.

Comparisons used a paired bootstrap over patients (2000 resamples), with Holm correction across the ten arms of the pooling sweep. Precision was computed by the Hanley–McNeil closed form [11], validated against a bootstrap; minimum sample size follows Riley *et al.* [12], applied only where its inputs are defined.

3 Results

3.1 The model does not exceed the chart

Table 2 gives every arm on identical patients and folds. Pooled out-of-fold *EGFR* AUC was 0.618 ± 0.025 across three seeds. Smoking status alone reached 0.794; the five-variable clinical model 0.764.

The co-primary endpoint was $\Delta\text{AUC} = -0.029 \pm 0.015$, with every seed’s confidence interval including zero. No fusion topology exceeded the clinical model: soft voting reached 0.748, cross-fitted stacking 0.735, hard voting 0.690. Adding radiomics significantly *harmed* the combination (-0.130 , $P = .02$). A logistic regression on frozen encoder features scored 0.617 ± 0.052 — indistinguishable from the full network to three decimals.

EGFR results are shown per arm in Figure 2 and with paired deltas in Figure 3. *KRAS* was at chance (0.489 ± 0.005); Gevaert *et al.* likewise report no significant *KRAS* model on this cohort [2].

3.2 The predictions are also uncalibrated

Calibration slope averaged 0.19 against a perfect 1.0, indicating predictions roughly five times too extreme. At a base rate of 26.1%, a model predicting the prevalence for every patient achieves a Brier score of 0.193; the model scored 0.236. **The probabilities are therefore worse than a constant**, with resolution — the component only a better model can improve — between 0.009 and 0.017. This is why no operating point is pre-specified, and why any future externally validated artifact must carry explicit recalibration.

3.3 On the subset that should carry it, performance is weaker

Every *EGFR*-mutant patient in this collection is an adenocarcinoma; the 17 squamous and 3 NSCLC-NOS patients are wild-type without exception. Including them makes 13% of the cohort classifiable from histology alone with no imaging. On the pre-specified primary analysis set — segmented adenocarcinoma, $n = 97$, 23 positives — AUC falls to 0.563. The same pattern appears in the literature: Yamazaki *et al.* report 0.774 across histologies but 0.687 restricted to adenocarcinoma [7].

3.4 Localisation generalises to an unseen cohort

On 420 held-out NSCLC-Radiomics patients (Figure 5), attention mass inside the tumour was 0.292 against 0.048 for the spatial shuffle — a lift of +0.244, beating the shuffle in 11 of 15 evaluations. Pointing reached 0.609 against 0.476 for the centre prior and 0.005 for a randomly initialised head, which returned a lift of exactly +0.000 at a peak-to-mean ratio of 1.01.

Two qualifications belong in the same breath. First, **the margin over the pre-specified fold-rate threshold is one fold:** 11/15 (73.3%) against a required 71.4%. Second, the failures are concentrated rather than scattered — one training run localised in 2 of 5 folds while the others reached 5/5 and 4/5, and that same run localises in 2/5 internally while carrying the *highest* classification AUC of the three. Localisation is therefore a property of most training runs rather than of the method.

Representative attention maps appear in Figure 6. These are the first six segmented patients of a held-out fold in index order, not selected on score.

3.5 The peritumoral region carries the signal, but not significantly

Because the classifier pools from the grounded region — the lesion itself — we asked whether the *EGFR* signal is located where the grounding loss supervises attention. Over ten seeds on frozen encoder features, through identical nested folds, mean-pooling the peritumoral ring scored 0.695 against 0.632 for the whole crop and 0.629 for the tumour interior. A size- and shape-matched ring at a *random* location scored 0.575, so the gain is location-specific rather than an artefact of pooling a thin annulus. Adding a peritumoral branch to the classifier moved the trained model from 0.618 to 0.675 ± 0.017 , improving in 3 of 3 seeds.

This does not survive correction, and we report it as suggestive rather than established. The sweep scores ten arms on the same folds; under a paired bootstrap over patients with Holm correction, the best arm has $P_{\text{raw}} = .03$ and $P_{\text{Holm}} = .28$. End-to-end the branch is worth +0.058 (95% CI -0.008 to $+0.128$, $P = .09$). The direction is consistent across frozen and trained settings and the location controls behave; the significance is absent.

It is also a replication rather than a discovery: peritumoral radiomics for *EGFR* has been reported at least six times since 2022, with optimal ring widths of 1, 2, 3, 4, 6 and 15 mm [5–10]. What is new here is the tension it exposes — the grounding objective supervises attention *onto* the lesion, and the classification signal appears to sit just outside it.

3.6 Analytic choice is worth more than the effect

On identical features and estimator, the radiomics arm moved **0.136 AUC** across six defensible preprocessing choices, and **0.066** between a flat and a nested cross-validation protocol. Both exceed the effect sizes this literature reports as findings.

3.7 More patients would not settle it

Riley’s criterion requires 414 patients for the five-variable clinical model alone and 42,362 for the 512-feature frozen probe; this cohort has 153. Neither arm is identifiable at this sample size, which is why “the probe matches the model” cannot be read as evidence that the architecture is redundant.

Precision, not power, binds any external test. At the observed prevalence a 95% confidence-interval half-width of ± 0.05 requires $n = 661$. The entire open world of *EGFR*-labelled CT holds approximately 327 cases with 64 positives, giving ± 0.154 — an interval spanning chance and the published 0.80s simultaneously.

4 Discussion

Two claims usually bundled together separate cleanly here, and separate differently. Supervised localisation transfers to a cohort from another country and scanner fleet. Genotype prediction does not clear a five-variable chart model on the cohort it was trained on.

The convergence with Rodríguez Sánchez *et al.* is the strongest argument that this is a property of the problem rather than of our sample. Their 1,646 patients and 11,473 lesions, with NGS labels and biopsy-anchored training, reproduce the interval we obtain from 153 patients. Ten times the patients did not move the ceiling.

Limitations. The internal cohort is single-institution and historical. The classification arm has no external validation; that is justified by the measured precision bound above rather than by circumstance, and is reported as a finding rather than deferred. Inter-rater variability of the segmentations is not measured, because the source collection distributes one contour per lesion; a mask perturbation sweep is reported instead and is explicitly not a substitute. One of three training runs largely fails to localise. Finally, H_1 is confirmatory of a prior observation on an independent patient set rather than a blind prospective test: the external cohort was on disk, and the metric’s behaviour on it partly known, when the threshold was set. The registered plan states this precisely, and we state it here rather than let a reader infer a stronger design than was run.

Conclusion. On this cohort, weakly-supervised localisation generalises across institutions while *EGFR* prediction from lesion-anchored CT does not exceed routinely available clinical variables. Reports of AUC 0.80–0.89 should be read against the analytic-choice sensitivity and precision bounds measured here.

Data and code availability

All code, the pre-specified analysis plan with its deviation log, the completed CLAIM checklist, and every archived run result — including negative results — are available at <https://github.com/sathvikloke/THEIA>. Every number in this manuscript is regenerated by `scripts/reproduce.sh` from archived out-of-fold predictions, requiring no imaging data and no GPU.

Imaging is redistributed by The Cancer Imaging Archive under its own terms and is not included in the repository. NSCLC-Radiogenomics is CC BY 3.0; NSCLC-Radiomics is CC BY-NC 3.0. Figure 6 shows NSCLC-Radiogenomics images only, consistent with that licence.

Figures



Figure 1: Participant flow. Counts and exclusion reasons are generated from the pipeline rather than transcribed, and reconcile exactly ($211 - 114 = 97$). The pre-specified primary analysis set is segmented adenocarcinoma.

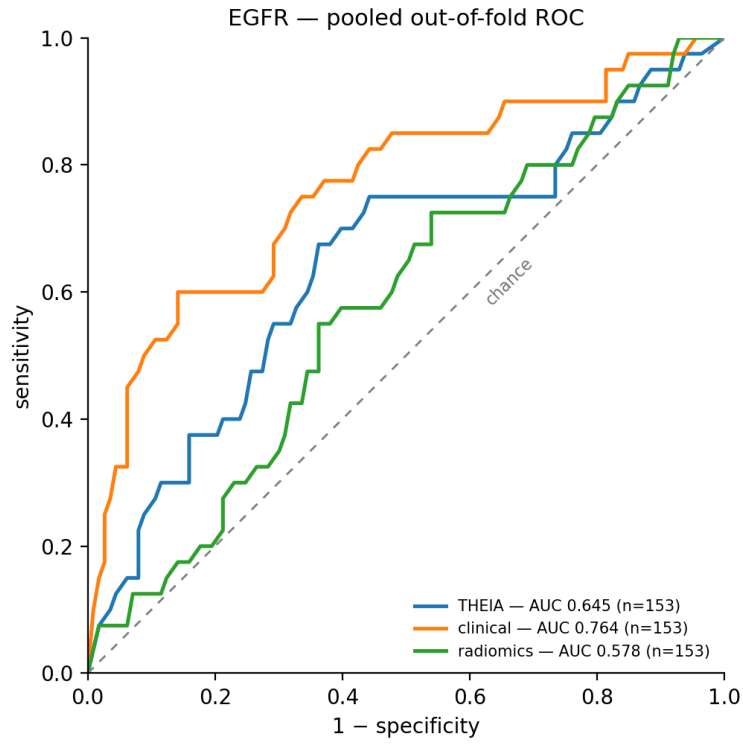


Figure 2: Pooled out-of-fold ROC by arm, on identical patients and folds. The chance line is drawn and labelled; each arm's n is printed, because arms do not all cover the same patients.

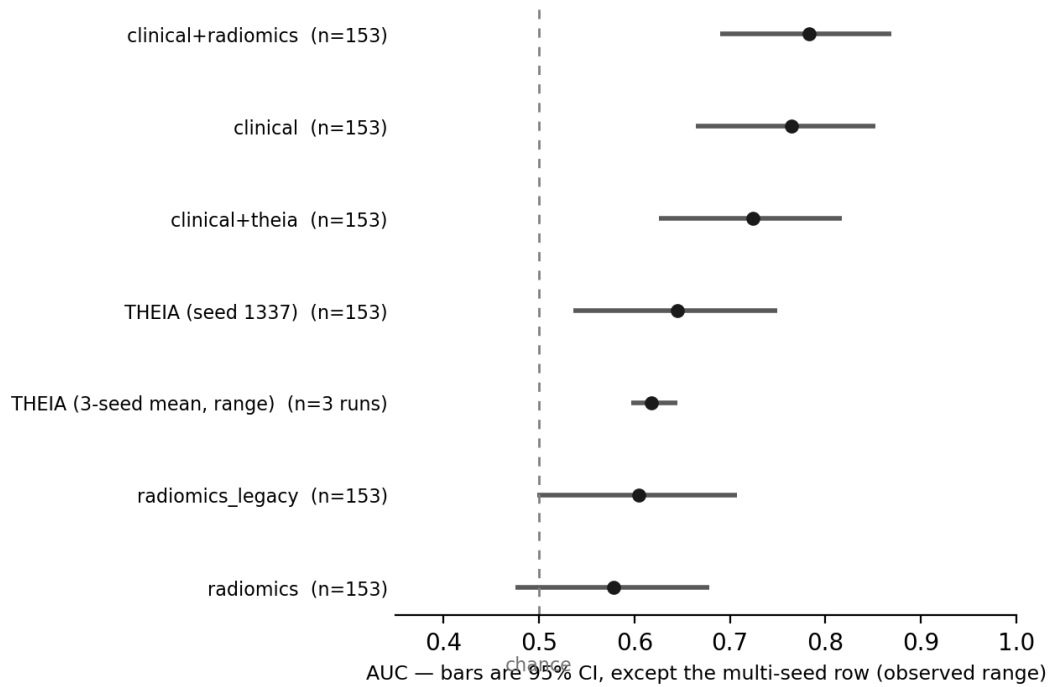


Figure 3: Every arm with its confidence interval. Overlapping intervals can still differ significantly when paired, which is why the paired deltas are reported alongside.

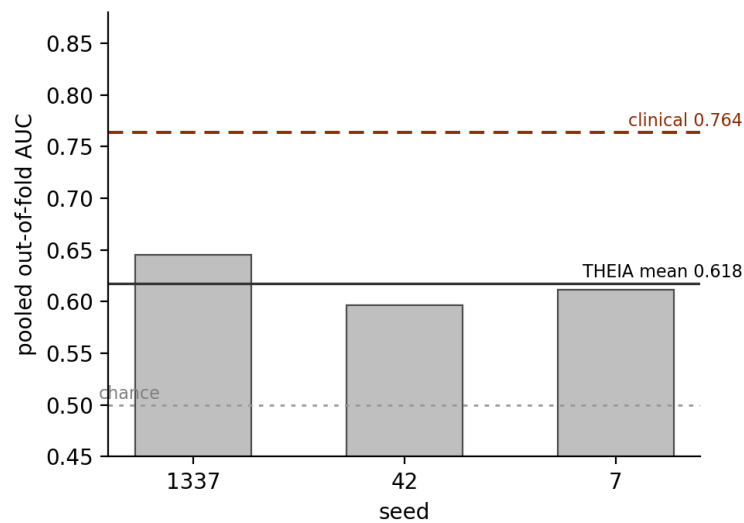


Figure 4: Per-seed AUC against the clinical baseline. The rerun spread is shown next to the effect being claimed; the gap to a five-variable chart model exceeds it.

External validation: 420 held-out NSCLC-Radiomics patients (Maastricht, NL) — mass lift +0.244, beats shuffle 11/15

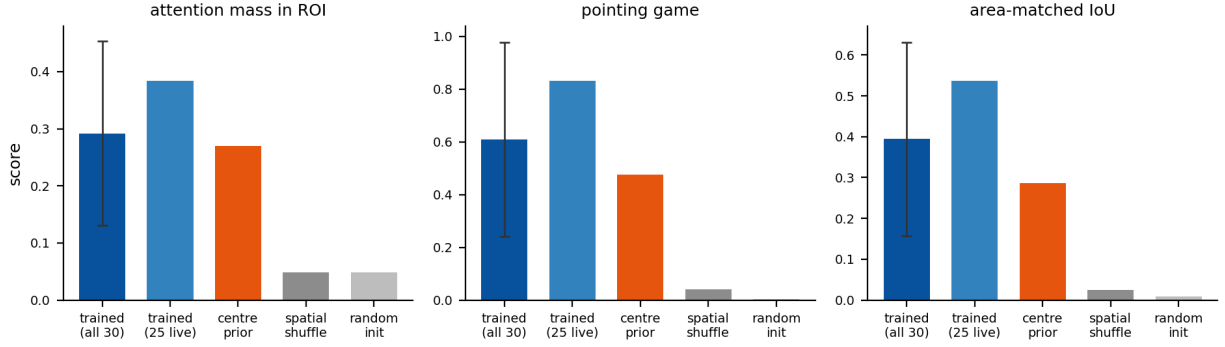


Figure 5: Primary endpoint: localisation on 420 held-out NSCLC-Radiomics patients, with all three controls in every panel. The centre prior is strong on attention mass by construction and weak on pointing and area-matched IoU, which are invariant to its width — so the trained model’s margin is read from the right-hand panels, not the left.

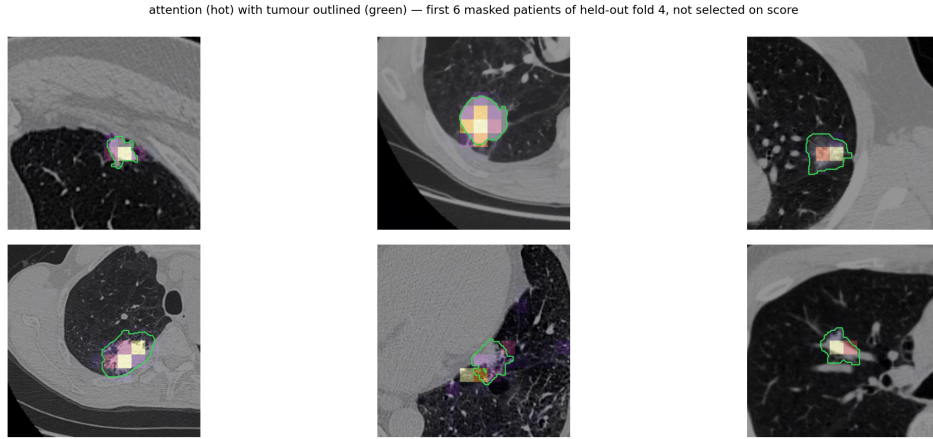


Figure 6: Attention maps with the tumour contour drawn from the reference segmentation. The first six segmented patients of a held-out fold in index order, *not* selected on score. Images from NSCLC-Radiogenomics (CC BY 3.0) [1].

Tables

Table 1: Cohort characteristics by *EGFR* status, restricted to the 153 patients with a known call. Never-smokers are 50% of mutants against 12% of wild-type; this is the textbook epidemiology and is most of what the clinical model knows.

Characteristic	<i>EGFR</i> mutant	Wild-type	All
<i>n</i>	40	113	153
Age, median [IQR]	68 [63–76]	69 [64–75]	69 [64–76]
Pack-years, median [IQR]	15 [9–34]	42 [26–56]	40 [20–50]
Female	22 (55%)	33 (29%)	55 (36%)
Male	18 (45%)	80 (71%)	98 (64%)
Never-smoker	20 (50%)	13 (12%)	33 (22%)
Former smoker	18 (45%)	78 (69%)	96 (63%)
Current smoker	2 (5%)	22 (19%)	24 (16%)
Asian	9 (22%)	6 (5%)	15 (10%)
Caucasian	13 (32%)	77 (68%)	90 (59%)
Other / not recorded	18 (45%)	30 (27%)	48 (31%)
Adenocarcinoma	40 (100%)	93 (82%)	133 (87%)
Squamous cell carcinoma	0 (0%)	17 (15%)	17 (11%)
NSCLC not otherwise specified	0 (0%)	3 (3%)	3 (2%)

Table 2: All arms and fusion topologies, on identical patients and folds. No combination of the imaging model with clinical variables exceeds the clinical model alone, and adding radiomics significantly harms it. Δ is against the clinical model; P from a paired bootstrap over patients.

Arm	<i>EGFR</i> AUC	Δ vs clinical	P
<i>Single arms</i>			
Smoking status alone	0.794	+0.030	—
Clinical (5 variables)	0.764	—	—
THEIA, 3 seeds	0.618 ± 0.025	−0.108	—
+ peritumoral branch	0.675 ± 0.017	−0.089	—
Frozen-feature logistic probe	0.617 ± 0.052	−0.147	—
Radiomics	0.578	−0.186	—
<i>Fusion topologies</i>			
Soft voting (THEIA + clinical)	0.748	−0.016	.59
Stacking (cross-fitted)	0.735	−0.029	.11
Soft voting (equal weights)	0.712	−0.052	.15
Soft voting (clinical + radiomics)	0.703	−0.061	.02
Hard voting	0.690	−0.074	.07
Soft voting (THEIA + radiomics)	0.634	−0.130	.02
<i>Co-primary endpoint</i>			
Δ AUC (clinical + THEIA) − clinical	−0.029 ± 0.015 , every seed’s CI includes zero		

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